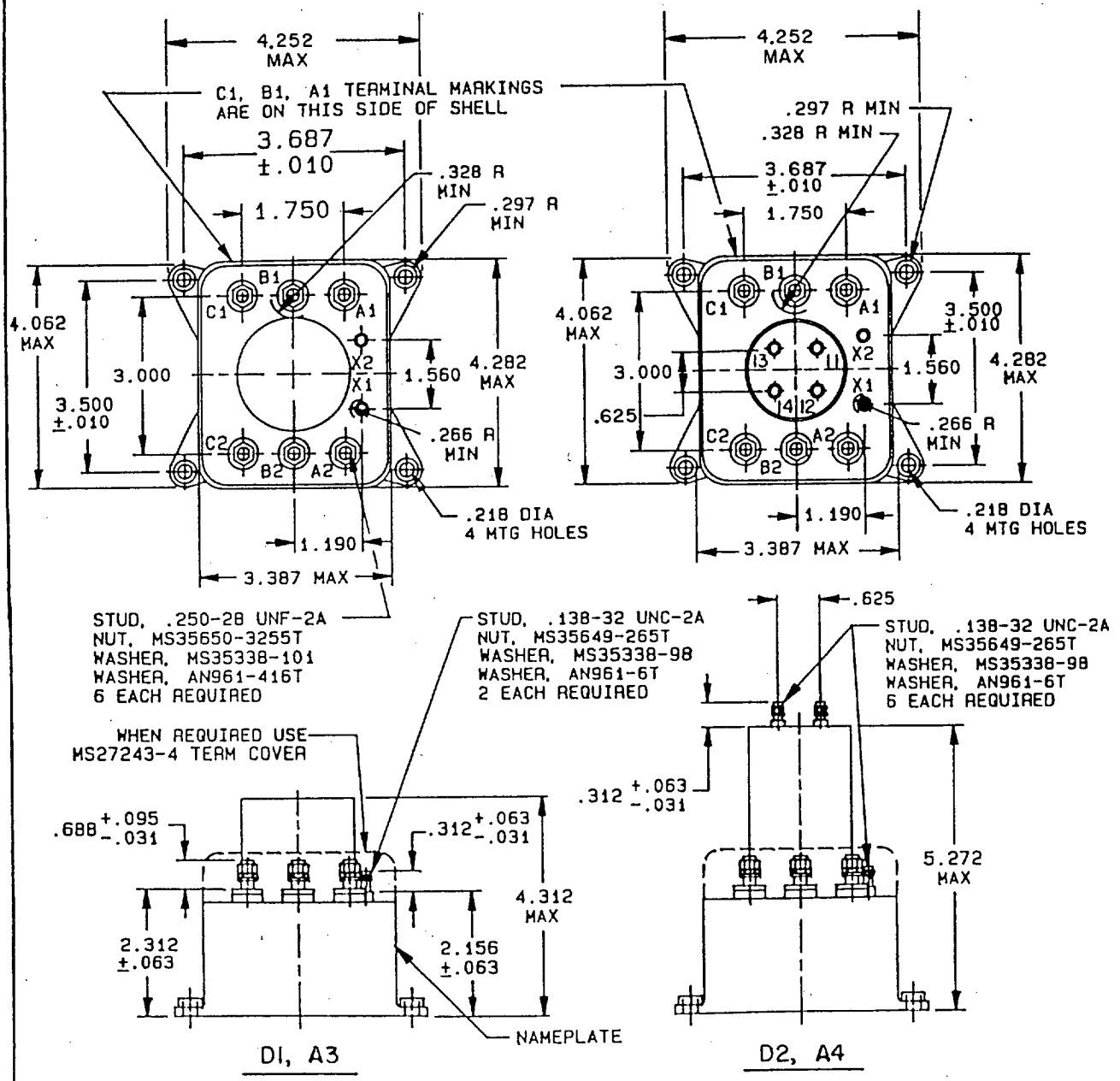
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NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

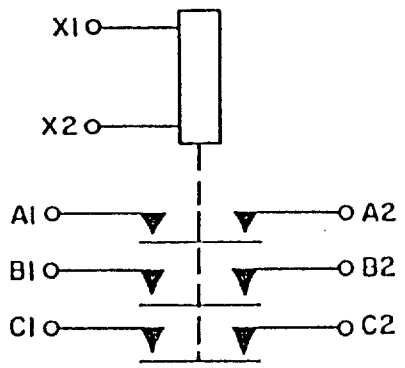
| | | |
|---|---|---|
| PREPARING ACTIVITY: Air Force - 85<br>CUSTODIANS: ARMY - NAVY - AS<br>AIR FORCE - DLA -<br>REVIEW: Navy-EC, Air Force-11,99<br>USER:<br>PROJECT NUMBER: 5945-0813-02 | MILITARY SPECIFICATION SHEET<br>TITLE RELAYS, ELECTROMAGNETIC, 100 AMPERES, 3 PST (N.O.), TYPE I, HERMETICALLY SEALED | SPECIFICATION SHEET NUMBER<br>MS24168P 25 Apr 90<br>SUPERSEDING<br>MS24168N 5 Nov 82<br>AMSC N/A FSC 5945 |
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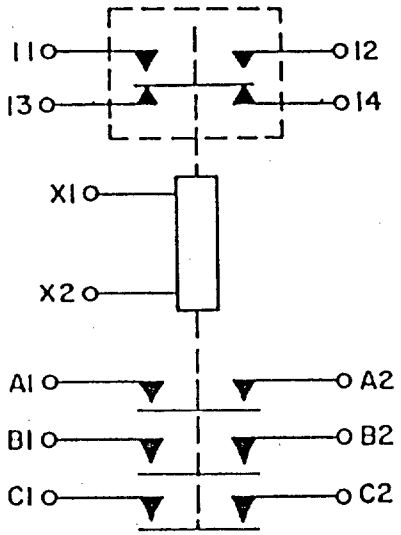
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CIRCUIT DIAGRAM A

(APPLICABLE TO RELAYS WITHOUT AUXILIARY CONTACTS)




CIRCUIT DIAGRAM B

(APPLICABLE TO RELAYS WITH AUXILIARY CONTACTS)

| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .010 | 0.25 | .312 | 7.92 | 2.312 | 58.72 |
| .031 | 0.79 | .328 | 8.33 | 3.000 | 76.20 |
| .063 | 1.60 | .625 | 15.88 | 3.387 | 86.03 |
| .095 | 2.41 | .688 | 17.48 | 3.500 | 88.90 |
| .138 | 3.51 | 1.190 | 30.23 | 3.687 | 93.65 |
| .218 | 5.54 | 1.560 | 39.62 | 4.062 | 103.17 |
| .250 | 6.35 | 1.718 | 43.64 | 4.282 | 108.76 |
| .266 | 6.76 | 1.750 | 44.45 | 4.312 | 109.52 |
| .297 | 7.54 | 2.156 | 54.76 | 5.272 | 133.91 |

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
4. Coil and auxiliary terminals may use additional flat washer for terminal seat.
5. Break all sharp edges and remove all burrs.
6. This specification sheet takes precedence over documents referenced herein.
7. Referenced documents shall be of the issue in effect on date of invitations for bid.

| | | |
|---|---|---|
| PREPARING ACTIVITY: Air Force - 85<br>CUSTODIANS: ARMY - NAVY - AS<br>AIR FORCE - DLA -<br>REVIEW: Navy-EC, Air Force-11,99<br>USER:<br>PROJECT NUMBER: 5945-0813-02 | MILITARY SPECIFICATION SHEET<br>TITLE RELAYS, ELECTROMAGNETIC, 100 AMPERES, 3 PST (N.O.), TYPE I, HERMETICALLY SEALED | SPECIFICATION SHEET NUMBER<br>MS24168P<br>SUPERSEDING<br>MS24168N<br>AMSC N/A FSC 5945 |
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TABLE I. Dash numbers and characteristics.

| Dash number MS24168- | Type | Coil type | Terminal type | Mounting or mating socket | Auxiliary contacts | Maximum weight in pounds 1/ |
|---|---|---|---|---|---|---|
| D1 | | dc | | | None | 3.4 |
| D2 | | dc | | | Yes | 3.6 |
| A3 | | ac | | | None | 3.8 |
| A4 | | ac | | | Yes | 3.8 |

1/ Weights include covers and barriers.

TABLE II. Operating characteristics.

| Dash number MS24168- | Coil data | | | | | | | | | | | Time - milliseconds max | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Nominal | | Max | Max pick-up voltage | | | | Drop out voltage 2/ 3/ | Hold voltage 2/ 3/ | Operate 4/ | Release 5/ | Contact bounce | | | |
| | | Volts 1/ | Frequency Hz | Res +15% -10 | Volts | Ampere | Normal 2/ | High temp test | Cont current test | | | | | Main | | Aux | |
| | | | | | | | | | | | | | | NO | NC | NO | NC |
| D1 | X1,X2 | 28 | dc | 55 | 29 | 0.6 | 18 | 21 | 22.5 | 1.5 | 7 | 60 | 25 | 2.5 | --- | --- | --- |
| D2 | X1,X2 | 28 | dc | 55 | 29 | 0.6 | 18 | 21 | 22.5 | 1.5 | 7 | 60 | 25 | 2.5 | --- | 4.5 | 4.5 |
| A3 | X1,X2 | 115 | 400/50/60 6/ | --- | 120 | --- | 90 | --- | --- | 10 | 40 | 65 | 80 | 5 | --- | --- | --- |
| A4 | X1,X2 | 115 | 400/50/60 6/ | --- | 120 | --- | 90 | --- | --- | 10 | 40 | 65 | 115 | 5 | --- | 7 | 7 |

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ At 50 and 60 Hz, chattering may occur at or near dropout voltage when voltage is slowly decreased.
4/ With nominal coil voltage.
5/ From nominal coil voltage.
6/ Coils will operate at either 50, 60, or 400 Hz when derated to +71°C maximum ambient temperature.

PREPARING ACTIVITY: CUSTODIANS: ARMY - Air Force, NAVY - AS, AIR FORCE, DLA
REVIEW: Navy-EC, Air Force-11,99
USER:
PROJECT NUMBER: 5945-0813-02
DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

TITLE: MILITARY SPECIFICATION SHEET
RELAYS, ELECTROMAGNETIC, 100 AMPERES, 3 PST (N.O.), TYPE I, HERMETICALLY SEALED

SPECIFICATION SHEET NUMBER: MS24168P
SUPERSEDING: MS24168N
AMSC N/A
FSC 5945

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TABLE III. Rated contact load (amperes per pole) case grounded.

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, 1-phase | | | | 115/200 V ac, 3-phase 1/ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | |
| | | NO | NC | NO | NC | 400 Hz | 50/60 Hz | 400 Hz | 50/60 Hz | 400 Hz | 50/60 Hz | 400 Hz | 50/60 Hz |
| Resistive | 50 | 100 | --- | 5 | 5 | 100 | 50 | 5 | 2 | 100 | 50 | --- | --- |
| Inductive | 10 | 100 | --- | 5 | 5 | --- | --- | --- | --- | --- | 50 | --- | --- |
| Inductive | | | | | | | | | | | | | |
| Motor | 50 | 100 | --- | --- | --- | 100 | 20 | --- | --- | 100 | 40 | --- | --- |
| Lamp | 50 | --- | --- | .75 | .75 | --- | --- | .75 | .75 | --- | --- | --- | --- |
| Transfer load 2/ | | | | | | | | | | | | | |
| Mechanical life reduced current | 100 | 25 | --- | 1.25 | 1.25 | 25 | 12.5 | 1.25 | 1.25 | 25 | 12.5 | --- | --- |
| Intermediate current | 50 | 10 | --- | Per spec | --- | 10 | 10 | --- | --- | 10 | 10 | --- | --- |

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Transfer load indicates relay suitability for transfer between unsynchronized ac power supplies at rating indicated.

PREPARING ACTIVITY: Air Force - 85
CUSTODIANS: ARMY - NAVY - AS; AIR FORCE - ; DLA -
REVIEW: Navy-EC, Air Force-11,99
USER:
PROJECT NUMBER: 5945-0813-02
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MILITARY SPECIFICATION SHEET
TITLE: RELAYS, ELECTROMAGNETIC, 100 AMPERES, 3 PST (N.O.), TYPE I, HERMETICALLY SEALED

AMSC N/A

SPECIFICATION SHEET NUMBER
MS24168P
SUPERSEDING MS24168N

FSC 5945



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## ELECTRICAL CHARACTERISTICS:

| | |
|---|---|
| Minimum insulation resistance, initial | 100 megohms |
| After life or environmental tests | 50 megohms |
| Dielectric strength (sea level) | 2-5 seconds |

| | Initial 1/ | | After life tests 1/ | |
|---|---|---|---|---|
| | 28 V dc | 115 V ac | 28 V dc | 115 V ac |
| Coil to case | 1,250 V rms | 1,500 | 1,000 V rms | 1,125 |
| Aux contacts | 1,250 V rms | 1,500 | 1,000 V rms | 1,125 |
| All other points | 1,250 V rms | 1,800 | 1,000 V rms | 1,350 |

Dielectric strength (altitude) 1 minute

| | 28 V dc 1/ | 115 V ac 1/ |
|---|---|---|
| Coil to case | 500 V rms | 500 |
| Aux contacts | 500 V rms | 500 |
| All other points | 700 V rms | 700 |

| | |
|---|---|
| Max contact drop initial | .150 volt |
| After life test | .175 volt |
| Overload current (NO) 2/ | 800 amperes |
| Rupture current (NO) 2/ | 1,000 amperes |
| Duty rating | Continuous |
| RFI specification | MIL-STD-461 |

(Applicable to coil circuits of ac operated relays)

1/ For A3 and A4, coil terminals X1 and X2 must be shorted together for all dielectric testing between coil to case, coil to main or auxiliary contacts, and coil to all other points.
2/ For 50/60 Hz, rupture and overload current (NO) 200 amperes.

## ENVIRONMENTAL CHARACTERISTICS:

| | |
|---|---|
| Temperature range 1/ | -70° to +125°C |
| Maximum altitude rating | 80,000 ft |
| Shock G-level | 25 g's |
| Duration | 6-9 ms |
| Max duration contact opening | 2 ms |
| Vibration - Sinusoidal (see table IV) | |
| G-level | G |
| Frequency range | Hz |
| Vibration - Random | N/A |
| Applicable specification | N/A |
| Power spectral density | N/A |
| RMS G minimum | N/A |
| Frequency range | N/A |
| Curve | N/A |
| High shock | N/A |
| Acceleration | 15 g's |

1/ Coils operated at either 50, 60, or 400 Hz, require ambient temperature not to exceed +71°C maximum.

TABLE IV. Vibration levels.

| Dash number MS24168- | 5-10 Hz | 10-55 Hz | 55-250 Hz | 250-500 Hz | 500-1,500 Hz |
|---|---|---|---|---|---|
| D1 | | | | 5 g's | 3 g's |
| D2 | | | | 4 g's | 3 g's |
| A1 | .08 DA | .06 DA | 10 g's | 5 g's | 3 g's |
| A2,A3,A4 | | | | 4 g's | 3 g's |

TABLE V. Qualification by similarity.

| Dash number MS24168- | Loads | | | | | | Dynamics 1/ | | | Environmental | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Type I | | | Type I ER | | | | | | | | |
| | A | B | C | D | E | F | A | X | Y | A | B | C |
| D1 | 4 | | | | | | 2 | | | 4 | | |
| D2 | 4 2/ | | | | | | 2 | | | 4 2/ | | |
| A3 | | | 1 | | | | | | 1 | | | 1 |
| A4 | | | 1 2/ | | | | | | 1 | | | 1 2/ |

1/ All relays must be tested. Reference MIL-R-6106, appendix I.
2/ Testing of relays with auxiliary contacts. Reference MIL-R-6106, appendix I.

PREPARING ACTIVITY: Air Force - 85
CUSTODIANS: ARMY - NAVY - AS AIR FORCE - DLA -
REVIEW: Navy-EC, Air Force-11,99
PROJECT NUMBER: 5945-0813-02
USER:
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SPECIFICATION SHEET NUMBER
MS24168P
SUPERSEDING MS24168N
AMSC N/A FSC 5945

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